YUANZHENG WEN

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EDUCATION

University of Iowa — Fall 2023- present (Expetced: Fall 2026)
Department of Physics and Astronomy — Ph.D. Candidate in Physics

University of Iowa — Dec 2025
Department of Physics and Astronomy — M.S. in Astronomy

Chengdu University of Technology — Jun 2022
Department of Geophysics and Space Sciences — B.S. in Space Sciences and Technology

PUBLICATIONS (CITATION: 65 H-INDEX:5)

1. Ionospheric TEC and plasma anomalies possibly associated with the 14 July 2019 Mw 7.2 Indonesia Laiwui earthquake, from analysis of GPS and CSES data

YZ Wen, D Tao, GX Wang et al.

Earth and Planetary Physics, doi: http://doi.org/10.26464/epp2022028

2. Statistical investigations of the radial IMF component on the current sheet structure in the Martian magnetotail: MAVEN observations

Y. Wen, Z. Rong, H. Nilsson et al.

The Astrophysical Journal, doi: https://doi.org/10.3847/1538-4357/ae11b3

3. Multi-Point Observations of the Magnetic Reconnection in the Martian Magnetotail Triggered by an Interplanetary Magnetic Field Rotation

Y. Wen, J. S. Halekas, H. W. Shen et al.

The Astrophysical Journal Letters , doi: https://doi.org/10.3847/2041-8213/adbf10

4. Magnetic Reconnection as a Potential Trigger for Magnetotail Flapping at Mars: Insights from MAVEN and Tianwen-1 Observations (selected as *EOS Research Spotlight* & Cover Article)

Y. Wen, J. S. Halekas, H. W. Shen et al.

AGU Advances, doi: https://doi.org/10.1029/2026AV002343

5. In situ evidence of a twisted loop–like magnetic flux rope at Mars: A MAVEN and Tianwen-1 Close Conjunction

Y. Wen, J. S. Halekas, T. Hara et al.

The Astrophysical Journal Letters, doi: https://doi.org/10.3847/2041-8213/ae9aa0

6. Magnetic reconnection in the Martian magnetotail: the first multi-point perspective

Y. Wen, J. S. Halekas, H. W. Shen et al. (In preparation)

7. Are the Significant Ionospheric Anomalies Associated with the 2007 Great Deep-Focus Undersea Jakarta-Java Earthquake?

D. Tao, G. Wang, J. Zong, **Y. Wen** et al.

Remote Sensing, doi: https://doi.org/10.3390/rs14092211

8. Statistical Analysis of Ion Properties in the Martian Magnetosheath Based on MAVEN Observations: A Comparison of Core and Total Populations

H. Shen, J. S. Halekas, S. M. Currcy, C. Zhang, **Y. Wen** et al.

The Astrophysical Journal, doi: https://dx.doi.org/10.3847/1538-4357/adf6a9

9. Strongly Sub-Isothermal Polytropic Behavior of Ions in the Martian Magnetosheath Revealed by MAVEN Observations

H. W. Shen, J. S. Halekas, S. Xu, **Y. Wen** et al.

The Astrophysical Journal, doi: https://doi.org/10.3847/1538-4357/ae5a91

RESEARCH EXPERIENCE

University of Iowa Jan 2024 - present
Graduate Research Assistant **Supervisor: Prof. Jasper Halekas**

- **Project: Exploring the Dynamic Magnetosphere of Mars Through Multi-point Approach**
- Revealed IMF rotation-triggered magnetic reconnection in the magnetotail of Mars
- Identified the magnetic reconnection as a potential triggering mechanism for tail flapping at Mars

LASP-University of Colorado, Boulder May 2022 - Aug 2023
Undergraduate Researcher **Supervisor: Prof. David Brain & Prof. Hans Nilsson**

- **Project: Joint Observations of Mars' Tail Ion Escape Evolution from MAVEN and MEX**
- Analyzed synchronized MAVEN STATIC and Mars Express ASPERA-3 plasma observation to study ion escape evolution in Mars' magnetotail
- **Presentation at ASEPRA Team Meeting by IRF**

Swedish Institute of Space Physics (IRF), Kiruna Apr 2021 - Aug 2022
Unddergraduate Researcher **Supervisor: Prof. Hans Nilsson & Prof. Mats Holmstrom**

- **Project: Solar Wind and Planetary Ions Mixing Investigations in the Vicinity of Martian Tail Region with MEX and MAVEN**
- Search and analysis of physical characters of ion mixing region near the Mars' tail boundary based on Mars Express and MAVEN observations

Institute of Geology and Geophysics, Chinese Academy of Sciences Jul 2021 - Oct 2021
Undergraduate Researcher **Supervisor: Prof. Zhaojin Rong**

- **Project: Statistical Investigations of the Flow-Aligned Component of IMF Impact on Magnetic Field Structure in Martian Magnetotail: MAVEN Observations**
- Analysis of magnetic field and plasma data from MAVEN to study IMF influences on Mars' magnetotail structure
- **Oral Presentation at 2022 AOGS Meeting in Singapore.**
- **First-author paper at *The Astrophysical Journal*.**

National Space Science Center, Chinese Academy of Sciences Jul 2020 - Sep 2020
Undergraduate Researcher **Supervisor: Dr. Yiteng Zhang**

- **Project: MHD Simulation of Mars Space Environment**
- Modeling the plasma environment of Mars and studied crustal magnetic fields influence on space environment
- **Supported by Chinese Academy of Sciences Research Fellowship.**

Chengdu University of Technology Sep 2019 - Dec 2020
Undergraduate Research Assistant **Supervisor: Prof. Dan Tao**

- **Project: Investigations of Seismic Ionospheric Disturbances with GPS and CSES**

- Analysis of global and local ionospheric anomalies before strong earthquakes based on satellite observations
- **First-author paper at *Earth and Planetary Physics*** and co-author paper at *Remote Sensing*

SELECTED HONORS AND AWARDS

Award for Outstanding Overseas Study Elite, China Scholarship Council ($6000)	2026
SHIELD Summer School Awardee, Boston University	2026
Ballard and Seashore Fellowship, University of Iowa ($11,653)	2026
NASA Heliophysics Summer School Awardee, UCAR	2025
AGU 2024 Outstanding Student Presentation Award (top 2%-5% of student presenters)	2025
Pfeiffer and Maltby Scholarship, University of Iowa ($1500)	2025
Undergraduate Research Fellowship, Chinese Academy of Sciences	Sep 2020/2021
Honorary Student of CAS-USTC International Summer School in Planetary Sciences	Aug 2020/2021
Honorary Student of Space Physics Summer School, ISPAT, Peking University	Jul 2021
National Scholarship, Ministry of Education of China	Sep 2020

CONFERENCES & TALKS

Y. Wen "Magnetic Reconnection in the Martian Magnetotail: A Multi-Point Perspective" (**Invited**) GEM Workshop 2026 , Portland, ME, Jul 2026

Y. Wen, J. S. Halekas, H. Shen, et al. "Magnetic Reconnection As Potential Drivers for Tail Flapping at Mars" AGU, New Orleans, LA, Dec 2025

Y. Wen, Z. Rong, H. Nilsson, et al. "Statistical Investigations of the Radial IMF Component Impact on the Magnetotail Current Sheet Structure of Mars: MAVEN Observations" AGU, New Orleans, LA, Dec 2025

Y. Wen, J. S. Halekas, H. Shen, et al. "Magnetic Reconnection associated Tail Flapping at Mars" MAVEN Project Science Groups Meeting, Boulder, CO, Sep 2025

Y. Wen, J. S. Halekas, H. Shen, et al. "Multi-Point Observations of the Dynamic Magnetotail of Mars: Exploring the Potential Mechanism for Magnetotail Current Sheet Flapping" MAVEN Project Science Groups Meeting, Morgantown, WV, May 2025

Y. Wen, J. S. Halekas, H. Shen, et al. "Multi-point Observations of Magnetic Reconnection in the Martian Magnetotail Triggered by an Interplanetary Magnetic Field Rotation" AGU, Washington DC, Dec 2024

Y. Wen, D. A. Brain, H. Nilsson, et al. "IMA-MAVEN comparison in the magnetotail of Mars" ASPERA Team Meeting at IRF, Sweden, Jun 2023

Y. Wen, Z. Rong, H. Nilsson, et al. "Statistical Investigations of the Flow-aligned Component of IMF Impact on the Current Sheet Structure in the Martian Magnetotail: MAVEN Observations" AOGS Singapore, Aug 2022

PROFESSIONAL SERVICE

AGU Advances, Publications, Reviewed one manuscript	2026

Journal of Geophysical Research: Space Physics, Publications, Reviewed one manuscript 2026

Astronomy & Astrophysics, Publications, Reviewed one manuscript 2026

Geophysical Research Letters, Publications, Reviewed one manuscript 2025

TEACHING EXPERIENCE

Teaching Assistant of PHYS 3811: Electricity and Magnetism	Aug 2024-Dec 2024
Teaching Assistant of PHYS 1511: College Physics I	Jan 2024-May 2024
Teaching Assistant of PHYS 3811: Electricity and Magnetism	Aug 2023-Dec 2023
Teaching Assistant of Mathematical Methods for Physics	Mar 2020-Jun 2020
Teaching Assistant of College Physics	Sep 2020-Jan 2021
Private Tutoring in Math, Physics and MATLAB Programming	

COMUPTER SKILLS

Programming	MATLAB, IDL (SPEDAS), Python (irfpy)
Operation System	Windows, Linux (Ubuntu)
Space Mission Instruments	MAVEN (STATIC, SWIA, MAG), Mars Express (ASPERA-3 IMA) CSES (LAP, PAP), Tianwen-1 (MOMAG)

REFERENCES

Prof. Jasper Halekas — Professor, University of Iowa
Ph.D. Advisor — **jasper-halekas@uiowa.edu**

Prof. David Brain — Professor, University of Colorado, Boulder
Ph.D. Committee Member — **david.brain@lasp.colorado.edu**

Prof. Han-Wen Shen — Assistant Professor, National Central University, Taiwan
Collaborator — **hwshen@jupiter.ss.ncu.edu.tw**

Dr. Yaxue Dong — Research Scientist, LASP
Collaborator — **yaxue.dong@lasp.colorado.edu**

Dr. Shaosui Xu — Associate Research Physicist, UC Berkeley
Collaborator — **shaosui.xu@berkeley.edu**

Prof. Hans Nilsson — Professor, Swedish Institute of Space Physics
Undergraduate Research Advisor — **hans.nilsson@irf.se**

Prof. Zhaojin Rong — Professor, Institute of Geology and Geophysics, CAS
Undergraduate Research Advisor — **rongzhaojin@mail.iggcas.ac.cn**

Prof. Dan Tao — Associate Professor, Chengdu University of Technology
Undergraduate Research Advisor — **dan.tao@cdut.edu.cn**